

2.1 Algorithms

Why this unit is taught the way it is

What this unit teaches, and where it ends

13 lessons: 11 with an online experience, 1 paper assessment and 1 bonus challenge. 155 activities in all, worth 361 marks.

By the end of it a pupil should be able to:

- know what is meant by the term abstraction al-101
- know some examples of abstraction al-101
- understand why abstraction is needed in computational thinking al-101
- know what is meant by problem decomposition al-102
- know the advantages of decomposition when programming al-102
- know how to produce a structure diagram al-102
- understand how to solve problems by applying algorithmic thinking al-103
- be able to identify the inputs, processes and outputs of a problem al-103
- know the flowchart symbols al-104
- know how to make a flowchart al-104
- understand how to construct a program from a flowchart al-104
- know what is meant by the term pseudocode al-105
- understand how to write pseudocode al-105
- understand the OCR reference language al-105
- understand the linear search algorithm al-106
- understand that it is inefficient but simple and needs no ordering al-106
- understand the binary search algorithm al-107
- know the condition a list must meet for a binary search to work al-107
- understand which searching algorithm is quicker al-107
- understand the bubble sort algorithm al-108
- be able to show the steps of a bubble sort al-108
- know the advantages and disadvantages of a bubble sort al-108
- understand the merge sort algorithm al-109
- understand the insertion sort algorithm al-109
- be able to compare the three sorting algorithms al-109
- know what a syntax error is al-110
- know what a logic error is al-110
- be able to identify errors in code and suggest fixes al-110
- know what a trace table is al-111
- understand how and why trace tables help with debugging al-111
- know how to complete a trace table al-111

Why the lessons are in this order

Each lesson's starter retrieves the one before it, so the order is the retrieval schedule as much as the teaching one. The chain for this unit:

LESSON	TEACHES	RETRIEVES
1 Abstraction studio	know what is meant by the term abstraction	nothing before it in this unit
2 Breakdown bay	know what is meant by problem decomposition	Abstraction studio
3 Algorithm workshop	understand how to solve problems by applying algorithmic thinking	Breakdown bay
4 Flowchart factory	know the flowchart symbols	Algorithm workshop
5 Pseudocode press	know what is meant by the term pseudocode	Flowchart factory
6 Search corridor	understand the linear search algorithm	Pseudocode press
7 Halving hall	understand the binary search algorithm	Search corridor
8 Bubble bay	understand the bubble sort algorithm	Halving hall
9 Split and insert	understand the merge sort algorithm	Bubble bay
10 Fault finder	know what a syntax error is	Split and insert
11 Trace room	know what a trace table is	Fault finder
12 End of topic test	assesses the whole unit	Trace room
13 Algorithm arena	stretches what is taught	End of topic test

The assessment in this unit is placed where it is so that the reflection sheet still has lessons left to send a pupil back to.

What pupils find hard here

None of this unit's walls names a common mistake in so many words. The misconceptions it works against are in the wrong options of its questions instead: every one of those is a plausible wrong idea rather than filler, and the lesson plans list them question by question.

Every multiple-choice and Predict activity in the unit — 96 of them — offers wrong answers that are the misconception, not filler. The split across those options at the hinge question is the cheapest diagnostic in the lesson.

Where the explaining, modelling and practising happen

PHASE	WHERE IT IS
Retrieve	The starter on the worksheet, before any device is opened. It is answered on paper and corrected before the lesson moves on.
Explain	The slides and the station walls carry the same explanation, so a teacher can teach from the front and a pupil can re-read it in the room.
Model	There is no worked example on screen in this unit: its activities are questions to answer rather than techniques to copy. The modelling is yours, from the slides, before anyone starts - and it is the part to protect when the lesson runs short.
Practise	127 guided activities, then 28 independent ones.
Check	The badge questions mark as the pupil goes; the hinge question is for the class at once.
Apply	0 challenge activities, and the exam practice on the worksheet.
Review	The exit question and the confidence columns. The confidence columns supplement the evidence; they do not stand in for it.

How the website, the slides and the worksheet divide the work

They are not three copies of the same thing.

- **The slides** are for the front of the room: the explanation and the model, before anyone is working alone.

- **The experience** is where a pupil reads, decides and is marked. It gives the immediate feedback a teacher cannot give thirty pupils at once.
- **The worksheet** is the record and the writing. The key fact in their own words, the station challenge, the exam practice. It is also what survives when the devices do not.

The rule that holds them together is **write first, then answer at the badge**. A pupil who taps first has turned the lesson into a quiz game, and the worksheet is how you can see at a glance who did.

Why a 360° room, and what it should produce

The room is not there to be impressive. It is there because this unit's content is a set of related parts that a pupil has to hold together, and putting one part on each wall makes the set visible and navigable: a pupil can turn back to station 2 while answering station 4. The evidence it should produce is on paper — 66 key facts and 66 station challenges in a pupil's own words — and in the score screen, which shows which station a pupil lost marks on rather than one number.

No claim is made here that immersion improves learning on its own. What the format does is make the structure of a topic physical and keep a pupil's attention on one part at a time; the teaching is in the explanation, the example and the practice, as it would be anywhere else.

Support, and how it is taken away

- The worked example is on the early activities of a station and not on the late ones. That fade is deliberate.
- The hint is a ladder asked for one rung at a time, and the top of it is still the technique on different data, never the answer.
- Read aloud is on every task.
- Prescribed values are boxed and in a monospace face as well as coloured, so they survive a black-and-white printout and a high-contrast mode.
- **Lower literacy and EAL:** pre-teach the vocabulary (Abstraction, Computational thinking, Decomposition, Structure diagram, Algorithm, Process, Flowchart, Decision, Pseudocode, Iteration, Linear search, Worst case). Accept the key fact spoken before it is written, and in two or three words.
- **SEND:** pace is not fixed. Progress saves after every answer and reopens at the earliest unfinished question, so a pupil may finish a station next lesson without losing anything.
- A headset is never required.

Checking, and what to do with what you find

Three checks, in this order: the starter (did last lesson stick), the hinge question (is the class ready for independent work), the exit question (did this lesson stick). The first two change what you do in the lesson; the third changes what you start the next one with.

When a hinge question splits the class, go back to the station it sits on and work its example again, aloud, before anyone goes on. When a test comes back, the reflection sheet names the topics to revisit and review mode on those lessons reopens exactly the questions a pupil got wrong.

Where the approach comes from

The shape of a lesson here — retrieve, explain, model, practise with support that fades, check, apply, review — is ordinary evidence-informed practice rather than anything this product invented. Useful starting points, none of them about Revise 360:

- Rosenshine, *Principles of Instruction* (American Educator, 2012) — the daily review, small steps, worked examples and checking for understanding that this structure follows.
- Sweller, van Merriënboer and Paas on cognitive load and the worked-example effect — why the example comes before the independent task and why the support is withdrawn gradually.
- Dunlosky et al., *Improving Students' Learning With Effective Learning Techniques* (2013) — the evidence for retrieval practice and spacing, which the starters and the review mode use.
- Wiliam, *Embedded Formative Assessment* — hinge questions and what to do with the answers.

What is not claimed. Nothing above is evidence that Revise 360 improves outcomes. No trial of this product has been run, and immersion is not claimed to improve learning by itself. The citations are for the teaching structure;

whether these resources work in your classroom is a question your own assessment answers.

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